

Strategic Hypothesis Testing Under Competition

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Abstract

Many approval systems evaluate evidence produced by strategic agents whose rewards depend on rival approvals. We study a principal who commits to one approval threshold, while each agent decides whether to enter and how much evidence to collect. Approved agents receive rewards that are diluted by the number of other approvals. In the symmetric known-quality model, the many-agent game reduces to a scalar fixed point in an effective reward. This reduction gives a competition-adjusted baseline-entry boundary: below it, null-quality agents do not enter and false approvals are zero. Away from that region, the symmetric equilibrium gives relations among mFDR, pFDR, and FDR and a hierarchy of FDR upper bounds. We then study a private-signal extension in which agents observe private signals and act on correct posterior beliefs about their own quality. The fixed-point reduction and the FDR bounds continue to hold, but zero false approvals need not. Numerical results illustrate how threshold choice, competition, FDR bounds, and private signals affect the principal’s design problem.

1 Introduction

Principals often evaluate evidence generated by strategic agents rather than by passive data sources. A regulator reviews submissions from firms seeking approval. A platform evaluates claims made by competing sellers. A scientific sponsor compares evidence from teams that decide whether and how aggressively to test ideas before reporting results. In these settings, agents care not only about passing the principal’s test, but also about how many rivals pass with them. This creates a basic policy problem: how should a principal choose an approval rule when the rule affects entry, evidence collection, and the composition of approved claims?

Prior work has studied principal-agent testing, strategic participation and beliefs, and strategic sample-size choice. What remains less understood is how these forces interact under direct competition. With multiple approved agents, the value of approval can be diluted by rival approvals. That payoff dilution changes which agents enter, how much evidence they collect, and how the principal should interpret FDR-type error rates.

The analysis restricts attention to a single-threshold policy class. The principal chooses one approval threshold α . Each agent has latent quality μ_i , chooses whether to enter, chooses a sample size if entering, and is approved if the resulting p -value passes the rule. Approved agents receive a reward that decreases with the total number of approvals. This captures a reduced-form version of competition for a shared market while keeping the principal’s policy instrument close to a standard statistical approval rule.

We first study the known-quality model, where agents know their own μ_i . In the symmetric approval-dilution model, competition enters through a single endogenous object: the effective reward $r_*(\alpha)$. Once this reward is known, each agent solves a single-agent strategic testing problem. The effective reward itself is pinned down by a fixed point, because an agent’s payoff depends on how

*Reproducible code and data are available in the GitHub repository.

many other agents are approved in equilibrium. This reduction separates two questions: for a fixed threshold, what agents do; and, anticipating those responses, which threshold the principal should choose.

The fixed-point reduction identifies a competition-adjusted baseline-entry boundary $\hat{\alpha}_{\text{comp}}$. Below this boundary, the baseline-quality agent does not enter, and by monotonicity no null-quality agent enters. Hence false approvals are zero. Away from that zero-false-approval region, we derive relations among the three FDR notions mFDR, pFDR, and FDR, together with a hierarchy of FDR upper bounds. The sharper bounds use the equilibrium null-entry rate and the selected sample sizes; the coarser bound uses only the effective reward and the minimum positive cost.

We then study a private-signal extension. Agents may instead observe private signals and act on correct posterior beliefs about their own quality. Under ordered posteriors, only sufficiently favorable signals lead to entry. The symmetric fixed-point structure and the relations among mFDR, pFDR, and FDR continue to hold. The zero-false-approval result, however, need not survive: a null-quality agent can receive a favorable signal and enter.

The numerical section is organized around the principal’s threshold-design problem. It shows how equilibrium rewards, entry, approvals, true approvals, and FDR change with the approval threshold; compares the FDR upper bounds; illustrates the effect of market size; and shows how the private-signal model approaches the known-quality benchmark as signals become more informative.

Contributions.

- (i) We introduce a competitive extension of strategic hypothesis testing in which agents choose entry and sample size while approved rewards are diluted by rival approvals.
- (ii) We derive a scalar fixed-point reduction for the symmetric known-quality model and use it to characterize a zero-false-approval region through the baseline-entry boundary $\hat{\alpha}_{\text{comp}}$.
- (iii) We derive relations among mFDR, pFDR, and FDR in symmetric equilibrium and prove a hierarchy of FDR upper bounds that the principal can use for threshold choice.
- (iv) We extend the model to private signals and correct posterior beliefs, showing that the fixed-point reduction and FDR bounds survive, while zero false approvals do not automatically survive.

Roadmap. Section 2 discusses related work. Section 3 introduces the model and principal objectives. Section 4 studies the symmetric known-quality model. Section 5 studies posterior self-belief. Section 6 presents the numerical results, and Section 7 concludes.

2 Related Work

The analysis is most closely related to three papers. Bates et al. [2022] study principal-agent testing when one strategic agent controls evidence production and the principal can shape incentives through mechanisms such as contracts or e-values. Their focus is mechanism design. Here the principal commits to one approval threshold, and the central issue is how competition among approved agents changes the induced equilibrium.

Bates et al. [2023] study population-level error control under strategic participation and make agents’ private beliefs about their own projects explicit. The private-signal extension below uses the same broad idea that an agent may act on a posterior belief rather than directly observing its own

quality. The difference is that the belief-based decision problem here is embedded in a competitive many-agent environment with payoff dilution.

Hossain et al. [2025] study the within-agent strategic testing problem most directly. In their model, an agent chooses whether to participate and, if participating, how much evidence to collect. We retain strategic entry and sample-size choice, but add multiple agents whose rewards are diluted by rival approvals.

3 Model

We consider a principal who evaluates submissions from m strategic agents. Agent i has latent quality $\mu_i \in [0, 1]$, with baseline quality μ_b . In the known-quality model, the agent observes μ_i before choosing whether to enter and, if entering, how much evidence to collect. The principal commits to an approval threshold α . The null is $H_{0,i} : \mu_i \leq \mu_b$ and the alternative is $H_{1,i} : \mu_i > \mu_b$. Section 5 relaxes the known-quality assumption by replacing direct observation of μ_i with a private signal and a correct posterior belief about the agent’s own quality.

3.1 Agent actions and evidence

If agent i enters and chooses sample size $n_i > 0$, the results below can be instantiated with different one-sided evidence models. In the Bernoulli/binomial case,

$$X_{i1}, \dots, X_{in_i} \sim \text{Bernoulli}(\mu_i),$$

and the agent uses the corresponding one-sided exact binomial test against μ_b . In the normal (Gaussian) case,

$$X_{i1}, \dots, X_{in_i} \sim \mathcal{N}(\mu_i, \sigma^2),$$

with known variance σ^2 , and the agent uses the corresponding one-sided Z -test against μ_b . In either case the principal approves when $p_i \leq \alpha$. The action $n_i = 0$ represents staying out. Let

$$\text{Pass}(\alpha, \mu, n)$$

denote the resulting approval probability for a quality- μ agent facing threshold α and sample size n , with $\text{Pass}(\alpha, \mu, 0) = 0$.

The agent’s cost takes the form

$$C_i(n_i) = \mathbf{1}\{n_i > 0\}(c_0 + cn_i),$$

where $c_0 > 0$ is a fixed entry cost and $c > 0$ is a marginal sampling cost. This matches the cost structure in the single-agent model of Hossain et al. [2025] and generates both entry and strategic sample-size choice. In Sections 4 and 5, we restrict attention to a finite sample-size menu

$$\mathcal{N} = \{0, n_{\min}, \dots, n_{\max}\},$$

so the minimum positive cost floor

$$C_{\min} = c_0 + cn_{\min}$$

is well defined.

The results in Section 4 use only structural properties of the induced approval probability rather than model-specific tail algebra. The required conditions are that for each positive sample size n :

- (i) $\text{Pass}(\alpha, \mu, n)$ is nondecreasing in μ ;
- (ii) $\text{Pass}(\alpha, \mu, n)$ is nondecreasing in α ;
- (iii) $\text{Pass}(\alpha, \mu, n) \leq \alpha$ under the null $\mu \leq \mu_b$;
- (iv) the within-agent payoff takes the affine form $r \text{Pass}(\alpha, \mu, n) - C(n)$.

Both concrete examples above satisfy these properties directly. The formal statements are therefore written entirely in terms of $\text{Pass}(\alpha, \mu, n)$, while the numerical section uses the Bernoulli/binomial example.

3.2 Competition through approval dilution

Let $A_i \in \{0, 1\}$ denote the approval indicator for agent i , and let

$$K = \sum_{j=1}^m A_j$$

be the number of approved agents. If agent i is approved, the reward is diluted by the total number of approvals:

$$W_i = Rg(K),$$

where $R > 0$ is the monopoly reward and g is decreasing with $g(1) = 1$. The baseline specification is

$$g(K) = K^{-\gamma}, \quad \gamma \geq 0,$$

so larger γ corresponds to stronger competition. When $\gamma = 0$, the model collapses to the monopoly benchmark with reward R regardless of how many other agents are approved.

One natural interpretation is a therapeutic-class approval market. Here m is the number of firms developing related products, R is the gross value of approval when a product has the field to itself, $g(K)$ is the residual revenue share when K products are approved into the same market, c_0 is the fixed startup cost of launching a study, and c is the marginal cost of collecting additional binary-outcome evidence such as patient-level trial data.

3.3 Principal objectives

The principal cares about both false approvals and missed discoveries. Let

$$V = \sum_{i=1}^m \mathbf{1}\{\mu_i \leq \mu_b\} A_i, \quad S = \sum_{i=1}^m A_i, \quad U = \sum_{i=1}^m \mathbf{1}\{\mu_i > \mu_b\} A_i.$$

We use the standard error metrics, with the convention that ratio-type error metrics are set to zero when their denominator is zero:

$$\begin{aligned} \text{mFDR} &= \begin{cases} \frac{\mathbb{E}[V]}{\mathbb{E}[S]}, & \mathbb{E}[S] > 0, \\ 0, & \mathbb{E}[S] = 0, \end{cases} \\ \text{pFDR} &= \begin{cases} \mathbb{E}\left[\frac{V}{S} \mid S > 0\right], & \mathbb{P}(S > 0) > 0, \\ 0, & \mathbb{P}(S > 0) = 0, \end{cases} \\ \text{FDR} &= \mathbb{E}\left[\frac{V}{\max(S, 1)}\right]. \end{aligned}$$

Thus if $S = 0$ almost surely, then

$$\text{mFDR} = \text{pFDR} = \text{FDR} = 0.$$

For policy design, we also use the linear loss

$$L(\alpha) = \lambda_{\text{FP}} \mathbb{E}[V] + \lambda_{\text{FN}} \mathbb{E}[T - U],$$

where T is the number of truly nonnull agents and $\lambda_{\text{FP}}, \lambda_{\text{FN}} \geq 0$ are principal weights on false approvals and false negatives. The linear loss can be read either as the principal’s objective or as a simplified social welfare objective for a planner who dislikes both approving low-quality projects and failing to approve high-quality ones. When the principal requires zero false approvals, the threshold-design problem becomes the constrained version of the same objective.

3.4 Equilibrium viewpoint

After the principal chooses α , agents play a simultaneous Bayesian game. The main equilibrium concept is a symmetric Bayes-Nash equilibrium: all agents use the same rule from private quality μ_i to the entry and sample-size decision. Symmetry does not mean that realized qualities are equal across agents or that all agents take the same action ex post. The principal then solves a Stackelberg problem against the induced equilibrium population. In the symmetric approval-dilution model, the key reduction is that the many-agent interaction can be represented through an effective reward $r_*(\alpha)$ determined within the model: each agent solves a single-agent strategic-testing problem at reward level $r_*(\alpha)$, while $r_*(\alpha)$ itself is pinned down by the expected payoff dilution generated by those equilibrium responses.

This separates the analysis into two layers. Holding α fixed, each agent chooses the privately optimal action given its quality μ_i . Thus the equilibrium strategy is a best-response rule rather than a policy imposed by the principal. Anticipating those responses, the principal chooses α : either to satisfy a zero-false-approval requirement, or to minimize the weighted loss $L(\alpha)$ defined above.

The next section develops this reduction for the symmetric competitive model and applies it to equilibrium screening, error control, and threshold design under competition.

4 Symmetric Competition with Known Qualities

4.1 Effective reward and the fixed-point reduction

In the symmetric approval-dilution model, each agent’s continuation payoff depends on other agents only through the number of rival approvals. This induces an effective reward determined within the model,

$$r_*(\alpha) = R \mathbb{E}[g(1 + Y)], \quad Y \sim \text{Binomial}(m - 1, \rho(\alpha, r_*(\alpha))),$$

where $\rho(\alpha, r)$ is the approval probability of a representative agent solving the single-agent strategic-testing problem at threshold α and reward level r . The many-agent problem is therefore reduced to a fixed point in r .

This reduction isolates the competitive channel. Relative to the monopoly benchmark, competition enters only through the downward pressure it places on the effective reward. Once $r_*(\alpha)$ is known, equilibrium entry, sample size, and approval behavior are inherited from the single-agent strategic-testing problem.

The results in this section use only the structural properties of the approval probability listed in Section 3: monotonicity in quality and threshold, one-sided validity under the null, and affine reward-based utility. In particular, the results are not tied to a specific tail formula. The Bernoulli/binomial

model and the one-sided normal (Gaussian) model with known variance are both concrete instances of the same argument.

This yields a two-layer design decomposition. First, fixing α , the equilibrium best-response rule determines which qualities enter, which sample sizes they choose, and how often they are approved in their own interest. Second, the principal chooses α knowing that equilibrium response rule. The competition-adjusted baseline-entry boundary $\hat{\alpha}_{\text{comp}}$ below will therefore serve two roles: it marks the left edge of the zero-false-approval region for a principal that insists on zero false approvals, and it is a reference boundary for a principal that instead minimizes the weighted loss $L(\alpha)$.

To turn the fixed-point reduction into an existence theorem, we add one regularity condition. The quality distribution is atomless, and for each fixed reward level the set of qualities at which two actions are exactly tied has measure zero. This rules out aggregate jumps caused by a positive-mass block of qualities switching actions simultaneously at the same reward. Throughout this section, the positive sample-size menu is finite, say

$$\mathcal{N}_+ = \{n_{\min}, \dots, n_{\max}\},$$

so

$$C_{\min} = c_0 + cn_{\min}$$

is the common minimum positive entry cost.

Proposition 1 (Fixed-point reduction, existence, and uniqueness). *Fix α , restrict attention to a finite sample-size menu, and adopt the tie-break rule that favors the action with the larger pass probability and then smaller sample size. Suppose the quality distribution is atomless and that for every fixed reward level r and every distinct pair of actions $n \neq n'$, the indifference set*

$$\{\mu : r \text{ Pass}(\alpha, \mu, n) - C(n) = r \text{ Pass}(\alpha, \mu, n') - C(n')\}$$

has measure zero. Define

$$\rho(\alpha, r) = \mathbb{E}_{\mu}[\text{Pass}(\alpha, \mu, \text{BR}(\alpha, r)(\mu))]$$

and

$$\Phi_{\alpha}(r) = R \mathbb{E}[g(1 + Y_r)], \quad Y_r \sim \text{Binomial}(m - 1, \rho(\alpha, r)).$$

Then Φ_{α} is continuous and nonincreasing on $[0, R]$. Hence there exists a unique fixed point

$$r_{\star}(\alpha) = \Phi_{\alpha}(r_{\star}(\alpha)).$$

If every agent uses $\text{BR}(\alpha, r_{\star}(\alpha))$, that profile is a symmetric equilibrium, and any symmetric equilibrium induces the same fixed point.

Proof sketch. For fixed (α, μ) , each sample size induces an affine utility line in the reward level r . Under the tie-break rule, the best-response pass probability is nondecreasing in r . Hence the aggregate approval probability $\rho(\alpha, r)$ is also nondecreasing. The atomless-quality and null-tie assumptions make $\rho(\alpha, r)$ continuous in r . Since the dilution function g is decreasing, larger approval rates make the expected continuation reward weakly smaller, which makes Φ_{α} nonincreasing. Continuity then gives a fixed point on $[0, R]$, and uniqueness follows because $\Phi_{\alpha}(r) - r$ is strictly decreasing. At that fixed point, a unilateral deviator faces exactly the continuation reward level $r_{\star}(\alpha)$ induced by the rivals' common strategy. Hence the best-response rule $\text{BR}(\alpha, r_{\star}(\alpha))$ leaves no profitable deviation. $\square$

4.2 Competition-adjusted baseline-entry boundary

The first consequence is a zero-false-approval result. Stronger competition lowers $r_*(\alpha)$, which raises the profitability threshold for entry. This motivates a competition-adjusted left boundary of the baseline-entry set and, below that boundary, zero false approvals rather than merely asymptotic control.

Definition 1 (Competition-adjusted baseline-entry boundary). *Define*

$$V^+(\alpha, \mu; r) = \max_{n>0} \{r \text{ Pass}(\alpha, \mu, n) - C(n)\},$$

and

$$\hat{\alpha}_{\text{comp}} = \inf \{ \alpha : V^+(\alpha, \mu_b; r_*(\alpha)) \geq 0 \},$$

with the convention $\inf \emptyset = +\infty$. Here $V^+(\alpha, \mu; r)$ is the optimized value over positive entry actions only. Since staying out yields utility 0 and pass probability 0, while the tie-break rule favors larger pass probability, the baseline-quality agent enters under the stated tie-break rule if and only if $V^+(\alpha, \mu_b; r_*(\alpha)) \geq 0$. Equivalently, $\hat{\alpha}_{\text{comp}}$ is the left boundary of the baseline-entry set.

Proposition 2 (Zero false approvals below $\hat{\alpha}_{\text{comp}}$). *Under the tie-break rule from Proposition 1, if*

$$\alpha < \hat{\alpha}_{\text{comp}},$$

then the baseline-quality agent stays out and every agent with quality $\mu \leq \mu_b$ also stays out. Consequently the false-approval count is almost surely zero, and by the zero-denominator convention from Section 3,

$$\text{mFDR}(\alpha) = \text{pFDR}(\alpha) = \text{FDR}(\alpha) = 0.$$

Proof sketch. If $\alpha < \hat{\alpha}_{\text{comp}}$, then by definition

$$V^+(\alpha, \mu_b; r_*(\alpha)) < 0,$$

so the baseline-quality agent strictly prefers staying out. For fixed $(\alpha, r_*(\alpha))$, the positive-entry value is nondecreasing in quality because the approval probability is nondecreasing in μ . Therefore any agent with lower quality $\mu \leq \mu_b$ also prefers staying out. That shuts down all null participation, hence all false approvals. $\square$

For the comparative statics below, let $\hat{\alpha}_{\text{comp}}(g, R)$ denote the baseline-entry boundary computed under dilution rule g and monopoly reward parameter R . When only one of these two primitives varies, we suppress the other argument. Also fix two dilution rules g_{lo} and g_{hi} with

$$g_{\text{lo}}(k) \leq g_{\text{hi}}(k) \quad \text{for every } k \in \{1, \dots, m\},$$

so g_{lo} represents stronger competition in the pointwise sense.

Proposition 3 (Comparative statics of $\hat{\alpha}_{\text{comp}}$). *Suppose the unique symmetric equilibrium exists for every α .*

(i) *If competition becomes stronger from g_{hi} to g_{lo} , then*

$$\hat{\alpha}_{\text{comp}}(g_{\text{lo}}) \geq \hat{\alpha}_{\text{comp}}(g_{\text{hi}}).$$

(ii) If the monopoly reward parameter increases from R_1 to $R_2 \geq R_1$, then

$$\hat{\alpha}_{\text{comp}}(R_2) \leq \hat{\alpha}_{\text{comp}}(R_1).$$

Proof sketch. For fixed reward level r , the positive-entry value $V^+(\alpha, \mu_b; r)$ is nondecreasing in α . Thus higher approval thresholds weakly favor positive entry at that fixed reward. Stronger competition lowers the fixed-point reward $r_*(\alpha)$, while larger monopoly reward raises it. Since entry is easier at larger rewards, the set of α values for which the baseline-quality agent enters shrinks under stronger competition and expands under larger rewards, giving the ordering of the corresponding left boundaries $\hat{\alpha}_{\text{comp}}$. $\square$

Appendix A gives the full comparison argument for Propositions 2 and 3.

The numerical results in Section 6 suggest a clear comparative-statics pattern:

- stronger competition pushes $\hat{\alpha}_{\text{comp}}$ upward,
- larger rewards pull $\hat{\alpha}_{\text{comp}}$ downward,
- larger fixed or marginal sampling costs also push $\hat{\alpha}_{\text{comp}}$ upward.

Thus $\hat{\alpha}_{\text{comp}}$ is a summary statistic for the principal’s design problem: it marks the left edge of the zero-false-approval region, beyond which baseline entry can occur. Any $\alpha < \hat{\alpha}_{\text{comp}}$ lies in the zero-false-approval region, so a principal with a zero-false-approval requirement must remain in that region. A principal who instead minimizes the weighted loss $L(\alpha)$ faces a broader design problem: the optimum may lie below, near, or above $\hat{\alpha}_{\text{comp}}$, depending on how the principal trades off false approvals against missed high-quality projects.

Market size and payoff dilution. The number of agents m affects the symmetric model only through the approval-dilution channel. Holding fixed the representative-agent approval probability ρ , increasing m shifts the rival approval count

$$Y \sim \text{Binomial}(m - 1, \rho)$$

to the right in first-order stochastic dominance. For a decreasing dilution law g , this lowers the expected continuation reward $\mathbb{E}[g(1 + Y)]$. Thus the same mechanism behind Proposition 3 suggests that larger markets tend to enlarge the zero-false-approval region and push $\hat{\alpha}_{\text{comp}}$ upward, while smaller markets do the opposite. We do not state a full theorem in m because changing market size also changes the fixed-point approval rate inside the equilibrium map. The small-market and large-market rows in Section 6 should therefore be read as illustrative rather than as a complete comparative-statics analysis.

4.3 FDR control under competition

The next result gives quantitative FDR bounds away from the zero-false-approval region. The coarse cost-based bound using $C_{\min}$ can be sharpened in two ways. First, false approvals are driven by null-quality agents that actually enter, not by all entrants. Second, different sample sizes have different null-validity ceilings. For a representative agent in symmetric equilibrium, let n_i denote the selected sample size and let A_i denote the approval indicator. Define

$$\rho(\alpha) = \mathbb{P}(A_i = 1), \quad \nu_n(\alpha) = \mathbb{P}(\mu_i \leq \mu_b, n_i = n), \quad \nu(\alpha) = \sum_{n>0} \nu_n(\alpha).$$

For each positive sample size n , also define its null-validity ceiling

$$\bar{\alpha}_n(\alpha) = \sup_{\mu \leq \mu_b} \text{Pass}(\alpha, \mu, n) = \text{Pass}(\alpha, \mu_b, n) \leq \alpha,$$

where the equality uses monotonicity in quality and the inequality uses one-sided null validity. The first ratio below is therefore the most granular bound in the theorem, the second one is a cleaner null-entry summary, and the third recovers the earlier cost-based bound using $C_{\min}$.

In the symmetric equilibrium model, the reduction can be pushed further. Since the null-quality indicators and approval indicators

$$(Z_i, A_i) = (\mathbf{1}\{\mu_i \leq \mu_b\}, A_i)$$

are i.i.d. under the common equilibrium strategy, mFDR, pFDR, and FDR satisfy

$$\text{pFDR} = \text{mFDR}, \quad \text{FDR} = \mathbb{P}(S > 0) \text{mFDR}.$$

This gives relations among mFDR, pFDR, and FDR rather than only an inequality. Proposition 2 already handles the zero-false-approval region below $\hat{\alpha}_{\text{comp}}$.

Theorem 1 (FDR control under competition in symmetric equilibrium). *Work in the finite sample-size menu model*

$$\mathcal{N} = \{0, n_{\min}, \dots, n_{\max}\}, \quad C_{\min} = c_0 + cn_{\min}.$$

Assume a common approval threshold α , assume that under the symmetric equilibrium the null-quality indicators and approval indicators

$$(Z_i, A_i) = (\mathbf{1}\{\mu_i \leq \mu_b\}, A_i)$$

are i.i.d. across agents, and use the zero-denominator convention from Section 3. Let $r_(\alpha)$ be the unique equilibrium effective reward. When $\rho(\alpha) = 0$, interpret the displayed ratios below as zero. Then*

$$\begin{aligned} \text{mFDR}(\alpha) &\leq \frac{\sum_{n>0} \bar{\alpha}_n(\alpha) \nu_n(\alpha)}{\rho(\alpha)} \leq \alpha \frac{\nu(\alpha)}{\rho(\alpha)} \leq \alpha \frac{r_*(\alpha)}{C_{\min}}. \\ \text{pFDR}(\alpha) &= \text{mFDR}(\alpha), \quad \text{FDR}(\alpha) = \mathbb{P}(S(\alpha) > 0) \text{mFDR}(\alpha). \end{aligned}$$

In particular,

$$\text{FDR}(\alpha) \leq \mathbb{P}(S(\alpha) > 0) \frac{\sum_{n>0} \bar{\alpha}_n(\alpha) \nu_n(\alpha)}{\rho(\alpha)} \leq \mathbb{P}(S(\alpha) > 0) \alpha \frac{\nu(\alpha)}{\rho(\alpha)} \leq \alpha \frac{r_*(\alpha)}{C_{\min}}.$$

If $\alpha < \hat{\alpha}_{\text{comp}}$, then

$$\text{mFDR}(\alpha) = \text{pFDR}(\alpha) = \text{FDR}(\alpha) = 0.$$

Proof sketch. The action-specific bound starts from the null-quality agents that choose each sample size: for sample size n , the null approval probability cannot exceed $\bar{\alpha}_n(\alpha)$. Summing over actions gives a bound for the false-approval probability. Replacing each $\bar{\alpha}_n(\alpha)$ by α gives the simpler null-entry bound. Finally, zero utility from staying out and the positive cost floor imply that total entry is bounded by $r_*(\alpha)/C_{\min}$ times approval probability, which recovers the coarser cost-based bound. Under symmetry, the joint pairs (Z_i, A_i) are i.i.d. Hence pFDR and mFDR coincide exactly, while FDR equals $\mathbb{P}(S(\alpha) > 0) \text{mFDR}(\alpha)$. When $\alpha < \hat{\alpha}_{\text{comp}}$, Proposition 2 already gives zero false approvals. $\square$

The theorem also gives a direct threshold-design problem once the equilibrium map $\alpha \mapsto r_\star(\alpha)$ is computed. A principal with a zero-false-approval requirement can restrict attention to $\alpha < \hat{\alpha}_{\text{comp}}$. A principal with a nonzero FDR target can instead solve, for example,

$$\max_{\alpha: \text{FDR}(\alpha) \leq q} \mathbb{E}[U(\alpha)] \quad \text{or} \quad \max_{\alpha: B(\alpha) \leq q} \mathbb{E}[U(\alpha)],$$

where q is a target such as 0.05 and $B(\alpha)$ is one of the FDR upper bounds from Theorem 1. The same one-dimensional search also covers the linear loss $L(\alpha) = \lambda_{\text{FP}}\mathbb{E}[V] + \lambda_{\text{FN}}\mathbb{E}[T - U]$ from Section 3.

What fails in general is a simple boundary rule outside the zero-false-approval region. For each fixed action n , the approval probability $\text{Pass}(\alpha, \mu, n)$ is nondecreasing in α , but the equilibrium quantities $\rho(\alpha)$, $\nu_n(\alpha)$, $\nu(\alpha)$, $r_\star(\alpha)$, and expected true approvals $\mathbb{E}[U(\alpha)]$ also move through best-response switching and payoff-dilution feedback. With a finite action menu, those switches can create local reversals. Accordingly, neither the computed FDR nor its upper bounds nor $\mathbb{E}[U(\alpha)]$ need be monotone in α , and feasible sets such as

$$\{\alpha : \text{FDR}(\alpha) \leq q\} \quad \text{or} \quad \{\alpha : B(\alpha) \leq q\}$$

need not be intervals. Outside $\alpha < \hat{\alpha}_{\text{comp}}$, the principal should therefore treat threshold choice as a one-dimensional search over the candidate α values rather than as a single-crossing boundary calculation.

Appendix A gives complete proofs of Proposition 1, Proposition 3, and Theorem 1, and makes explicit which single-agent assumptions and payoff comparisons are used in the many-agent equilibrium analysis.

5 Private Signals and Self-Belief

The symmetric model above assumes that each agent observes its own quality exactly before choosing whether to enter and how much evidence to collect. In many applications, however, an agent only has private preliminary information about its own quality. We therefore consider a structural extension in which agent i privately observes a signal S_i taking values in an ordered space $\mathcal{S}$ and forms the correct posterior belief

$$\Pi_s = \text{Law}(\mu_i \mid S_i = s).$$

Here “belief” means a distribution over the agent’s own latent quality after seeing private pre-entry information. It is different from the population quality distribution across agents and from the principal’s prior over quality.

Fix a threshold α and an effective reward r . If the agent chooses action $a \in \{0\} \cup \mathcal{N}$, where $a = 0$ means staying out and $a \in \mathcal{N}$ is a positive evidence-collection action, its posterior expected utility is

$$U_{\text{SB}}(\alpha, r, a; s) = r \mathbb{E}_{\mu \sim \Pi_s}[\text{Pass}(\alpha, \mu, a)] - C(a).$$

The corresponding best response is

$$\text{BR}_{\text{SB}}(\alpha, r)(s) \in \arg \max_{a \in \{0\} \cup \mathcal{N}} U_{\text{SB}}(\alpha, r, a; s).$$

Thus the strategy is now a rule from private signal, or equivalently posterior belief, to action.

This extension separates two ideas that are often conflated. *Correctness* means the agent uses the true posterior Π_s . *Accuracy* means the signal structure is informative enough that the posterior is

tightly concentrated near the realized quality. The first concept is needed for the error-control results below; the second will matter when we compare the model back to the known-quality benchmark.

A canonical concrete signal specification, especially natural under the Bernoulli/binomial approval model, is a Beta-Binomial pilot stage. Suppose the realized qualities in the market are drawn from a population quality distribution $\mu_i \sim \text{Beta}(a, b)$. Before choosing whether to enter the formal approval process, agent i privately observes pilot evidence

$$Y_i \mid \mu_i \sim \text{Binomial}(k, \mu_i).$$

Then the correct posterior belief is

$$\Pi_y = \text{Law}(\mu_i \mid Y_i = y) = \text{Beta}(a + y, b + k - y).$$

In that concrete example, “belief” means exactly this posterior distribution, while “accuracy” is represented by the pilot sample size k : larger k gives more informative private signals and more concentrated posteriors. We keep the theorem statements below abstract because they rely only on the induced posterior ordering and posterior concentration, not on the conjugate Beta-Binomial form itself.

Proposition 4 (Signal-monotone entry under self-belief). *Suppose the signal space $\mathcal{S}$ is ordered and more favorable private signals induce more optimistic posteriors in the sense that if $s' \geq s$, then $\Pi_{s'}$ first-order stochastically dominates Π_s . Then for each fixed (α, r, a) the posterior utility*

$$s \mapsto U_{\text{SB}}(\alpha, r, a; s)$$

is nondecreasing. Consequently, the optimized posterior value

$$V_{\text{SB}}(\alpha, r; s) = \max_{a \in \{0\} \cup \mathcal{N}} U_{\text{SB}}(\alpha, r, a; s)$$

is nondecreasing, and the entry set

$$\{s \in \mathcal{S} : \text{BR}_{\text{SB}}(\alpha, r)(s) > 0\}$$

is an upper set. If $\mathcal{S}$ is finite, or if $\mathcal{S}$ is an interval with enough regularity, this upper set can be represented by a cutoff signal.

Proof sketch. For each positive action a , the map $\mu \mapsto \text{Pass}(\alpha, \mu, a)$ is nondecreasing by the structural conditions from Section 3. Thus first-order stochastic dominance of the posterior family implies that posterior expected pass probabilities are nondecreasing in the signal. Multiplying by the nonnegative reward r and subtracting the fixed action cost preserves the ordering. Taking the maximum over actions preserves it again. $\square$

The many-agent symmetric extension remains tractable. Assume the tuples (μ_i, S_i, X_i) are i.i.d. across agents, where X_i denotes the formal trial randomness after entry. Define the representative-agent approval probability

$$\rho_{\text{SB}}(\alpha, r) = \mathbb{E}[\text{Pass}(\alpha, \mu_i, \text{BR}_{\text{SB}}(\alpha, r)(S_i))].$$

Theorem 2 (Self-belief equilibrium and error control). *Assume the same structural conditions on the approval probability as in Section 3, work with the finite sample-size menu*

$$\mathcal{N} = \{0, n_{\min}, \dots, n_{\max}\}, \quad C_{\min} = c_0 + cn_{\min},$$

use the zero-denominator convention from Section 3, adopt the tie-break rule that favors the action with larger posterior expected pass probability and then the smaller positive action, and suppose the dilution function g is decreasing. Also suppose the signal distribution is atomless and that for every fixed reward level r and every distinct pair of actions $a \neq a'$ in $\{0\} \cup \mathcal{N}$, the indifference set

$$\{s : U_{\text{SB}}(\alpha, r, a; s) = U_{\text{SB}}(\alpha, r, a'; s)\}$$

has measure zero under the signal law. In the symmetric self-belief model with correct posteriors and i.i.d. tuples (μ_i, S_i, X_i) :

(i) the many-agent equilibrium again reduces to a unique scalar fixed point

$$r_{\text{SB}}^*(\alpha) = R \mathbb{E}[g(1 + Y)], \quad Y \sim \text{Binomial}(m - 1, \rho_{\text{SB}}(\alpha, r_{\text{SB}}^*(\alpha)));$$

(ii) the same FDR-bound hierarchy continues to hold. Writing

$$\rho_{\text{SB}}(\alpha) = \rho_{\text{SB}}(\alpha, r_{\text{SB}}^*(\alpha)),$$

$$\nu_{\text{SB},n}(\alpha) = \mathbb{P}(\mu_i \leq \mu_b, \text{BR}_{\text{SB}}(\alpha, r_{\text{SB}}^*(\alpha))(S_i) = n), \quad \nu_{\text{SB}}(\alpha) = \sum_{n>0} \nu_{\text{SB},n}(\alpha),$$

and, for each $n > 0$,

$$\bar{\alpha}_n(\alpha) = \sup_{\mu \leq \mu_b} \text{Pass}(\alpha, \mu, n) = \text{Pass}(\alpha, \mu_b, n) \leq \alpha,$$

and interpreting the displayed ratios as zero when $\rho_{\text{SB}}(\alpha) = 0$, we have

$$\text{mFDR}(\alpha) \leq \frac{\sum_{n>0} \bar{\alpha}_n(\alpha) \nu_{\text{SB},n}(\alpha)}{\rho_{\text{SB}}(\alpha)} \leq \alpha \frac{\nu_{\text{SB}}(\alpha)}{\rho_{\text{SB}}(\alpha)} \leq \alpha \frac{r_{\text{SB}}^*(\alpha)}{C_{\min}};$$

(iii) the same relations among mFDR, pFDR, and FDR continue to hold,

$$\text{pFDR}(\alpha) = \text{mFDR}(\alpha), \quad \text{FDR}(\alpha) = \mathbb{P}(S(\alpha) > 0) \text{mFDR}(\alpha).$$

Proof sketch. For a fixed signal value s , each action still generates an affine utility line in the reward level r , with slope equal to the posterior expected pass probability under that action. Under the same tie-break logic as in Section 4, the chosen posterior expected pass probability is therefore nondecreasing in r , which makes $\rho_{\text{SB}}(\alpha, r)$ nondecreasing. The atomless-signal and no-tie assumptions make $\rho_{\text{SB}}(\alpha, r)$ continuous in r . The same intermediate-value argument as in Section 4 gives existence of a fixed point and strict monotonicity of $\Phi_{\text{SB},\alpha}(r) - r$ gives uniqueness. The error-control proof then proceeds in the same hierarchy as Section 4: action-specific null validity yields the sharpest bound, summing over positive actions yields the simpler null-entry bound, and correct posterior utility yields the coarser cost-based bound through nonnegative conditional expected profit. Finally, under the i.i.d. self-belief environment, the null-quality indicators and approval indicators

$$(Z_i, A_i) = (\mathbf{1}\{\mu_i \leq \mu_b\}, A_i)$$

are i.i.d. Hence the relations among mFDR, pFDR, and FDR continue to hold. $\square$

The key difference from the known-quality model is that zero false approvals do not follow automatically. A null-quality agent may receive a favorable private signal and therefore enter. The sharper bound hierarchy and the pFDR/mFDR/FDR relations continue to hold, but the zero-false-approval conclusion below $\hat{\alpha}_{\text{comp}}$ does not automatically carry over.

The natural measure of belief accuracy in this abstract model is *informativeness* of the signal structure. To formalize that idea, let the signal structure be indexed by η , with more informative structures giving posteriors that are more concentrated at the realized quality. Write S_i^η for the private signal and

$$\Pi_{\eta,s} = \text{Law}(\mu_i \mid S_i^\eta = s)$$

for the corresponding correct posterior.

Proposition 5 (Posterior concentration recovers the known-quality benchmark). *Suppose that for each fixed (α, a) , the map $\mu \mapsto \text{Pass}(\alpha, \mu, a)$ is continuous, and suppose the posterior beliefs concentrate at the realized quality in the sense that*

$$\Pi_{\eta, S_i^\eta} \Rightarrow \delta_{\mu_i} \quad \text{almost surely as } \eta \rightarrow \infty.$$

Then for each fixed action a the posterior utility converges almost surely to its known-quality counterpart,

$$U_{\text{SB}}^\eta(\alpha, r, a; S_i^\eta) \rightarrow r \text{Pass}(\alpha, \mu_i, a) - C(a).$$

Consequently, away from known-quality indifference points and the known-quality entry boundary, posterior-based best responses and entry decisions converge almost surely to the known-quality benchmark.

Proof sketch. Weak convergence of the posterior to the point mass at μ_i and continuity of the approval probability imply convergence of posterior expected pass probabilities. Since the action menu is finite, the corresponding value functions converge as well. A positive optimality gap in the known-quality benchmark then guarantees eventual agreement of best responses and entry decisions. $\square$

This section therefore gives a private-signal generalization of the main analysis. The Beta-Binomial pilot stage above is only a convenient concrete signal specification, and the self-belief extension shows that most of the equilibrium and error-control logic does not actually rely on agents knowing their own quality exactly. What the known-quality benchmark delivers is the zero-false-approval conclusion, not the scalar fixed point or the relations among mFDR, pFDR, and FDR.

6 Numerical Results

This section has two numerical goals. First, it evaluates the equilibrium quantities from Sections 4–5 in one baseline parameter setting. Second, it turns those equilibrium quantities into the principal’s threshold-choice problem. The main text therefore emphasizes quantities directly relevant to threshold design: equilibrium reward, participation, true approvals, FDR, and the sharper FDR bounds.

6.1 Parameter setting and computation

Unless otherwise stated, the baseline parameter setting is

$$m = 20, \quad \mu_b = 0.5, \quad q = \text{Beta}(6, 5), \quad R = 30,$$

with positive sample sizes

$$n \in \{10, \dots, 60\}, \quad c_0 = 0.5, \quad c = 0.05,$$

power-law payoff dilution

$$g(K) = K^{-1},$$

and exact binomial tails for the approval rule. The plots over approval thresholds in Figure 2 use the grid

$$\alpha \in \{0.01, 0.02, \dots, 0.40\},$$

while the market-size frontier in Figure 3 extends this grid to 0.95 to display the tradeoff once false approvals are positive. Figure 4 uses the grid $\alpha \in \{0.01, 0.02, \dots, 0.99\}$ for the market-size design calculation. The comparative-statics calculations for $\hat{\alpha}_{\text{comp}}$ use the finer grid

$$\alpha \in [0.10, 0.80].$$

For each threshold, the symmetric known-quality model solves the unique fixed point from Proposition 1; the self-belief calculations use the same economic parameters together with a discrete Beta-Binomial pilot signal and a direct reward-grid fixed-point selection rule for the stepwise self-belief map.

6.2 Strategic evidence choice and threshold variation

Figure 1 shows the equilibrium sample-size policy $n^*(\mu)$ below and just beyond the baseline-entry boundary. At $\alpha = 0.20$, all agents with $\mu \leq \mu_b$ stay out. At $\alpha = 0.28$, the baseline-quality agent enters and the zero-false-approval conclusion from Proposition 2 fails. In both panels, the chosen sample size is nonmonotone in μ : once pass probability is already close to one, the marginal value of extra evidence becomes small and high-quality agents often step back to smaller experiments.

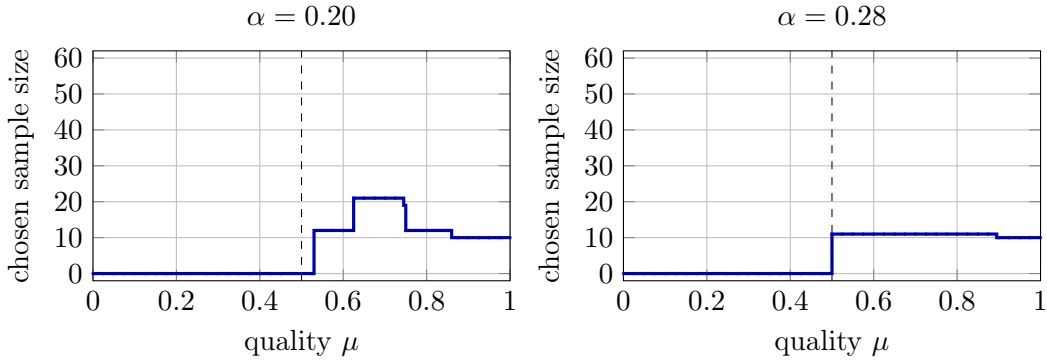


Figure 1: Equilibrium evidence-collection policies in the baseline symmetric parameter setting. The left panel shows the policy at $\alpha = 0.20$, and the right panel shows the policy at $\alpha = 0.28$. In both panels, the dashed line marks $\mu_b = 0.5$. At $\alpha = 0.20$, all null-quality agents stay out; at $\alpha = 0.28$, the baseline-quality agent enters and false approvals appear. The policy is computed on a quality grid with spacing 0.005 and is shown with small markers at sampled qualities. Because the action menu is discrete, the display is naturally step-like; the apparent breakpoints should be read at grid resolution rather than as exact claims about where the policy switches.

Figure 2 reports threshold-indexed quantities relevant to the principal's choice. The dashed line marks the threshold-grid estimate $\hat{\alpha}_{\text{comp}} \approx 0.28$. Equilibrium reward falls overall as the threshold

loosens and more rivals are approved, but the quantities relevant to the principal are not monotone in α . Entry, approval, true approvals, and the error curves all exhibit local reversals around $\alpha = 0.28$ – 0.30 and again near $\alpha = 0.39$. Those reversals come from discrete changes in equilibrium sample-size choice together with the induced movement in the fixed-point reward. Thus outside the zero-false-approval region $\{\alpha : \alpha < \hat{\alpha}_{\text{comp}}\}$, the principal’s threshold choice is a one-dimensional search problem rather than a boundary calculation.

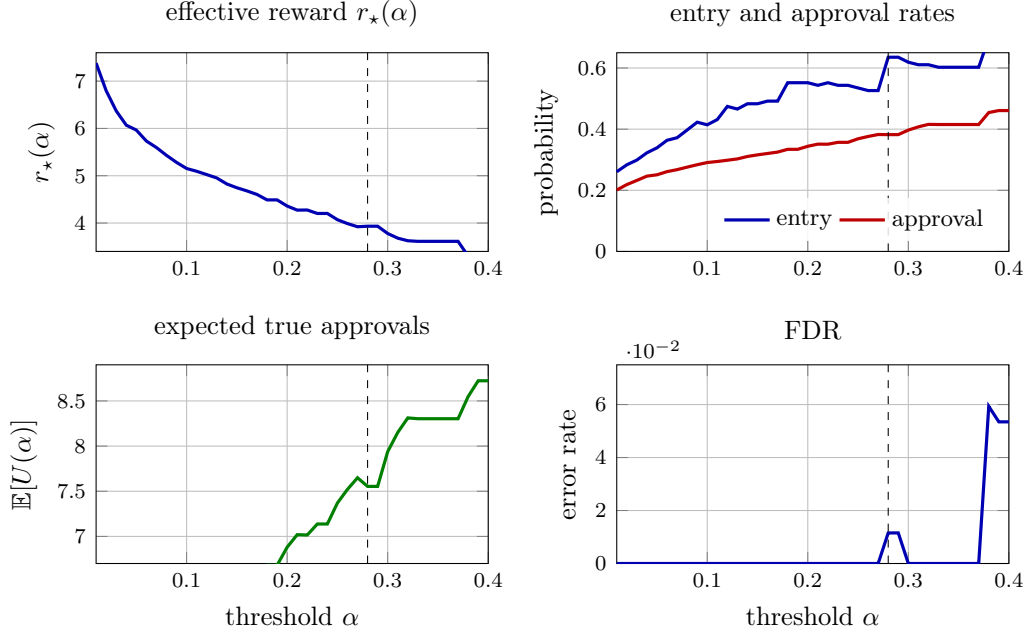


Figure 2: Threshold-indexed equilibrium quantities in the baseline parameter setting. The dashed vertical lines mark the threshold-grid estimate $\hat{\alpha}_{\text{comp}} \approx 0.28$. The upper-left panel plots the effective reward $r_*(\alpha)$; the upper-right panel plots the entry and approval probabilities; the lower-left panel plots expected true approvals $\mathbb{E}[U(\alpha)]$; and the lower-right panel plots FDR. The local reversals near $\alpha = 0.28$ – 0.30 and $\alpha = 0.39$ show why threshold choice under mixed FDR and power objectives cannot generally be reduced to a single crossing point.

Figure 3 presents the threshold-design problem as an error-rate frontier for three market sizes. Each point on a curve corresponds to one threshold on a wider grid, $\alpha \in \{0.01, 0.02, \dots, 0.95\}$. The horizontal axis is computed FDR, and the vertical axis is the false negative rate $1 - \text{sensitivity}(\alpha)$, where $\text{sensitivity}(\alpha) = \mathbb{E}[U(\alpha)]/\mathbb{E}[T]$ is the fraction of truly nonnull agents who are approved in expectation. The vertical stacks at $\text{FDR} = 0$ record thresholds for which false approvals are zero. Moving from $m = 1$ to larger markets changes the fixed-point reward through payoff dilution and changes which thresholds induce null entry.

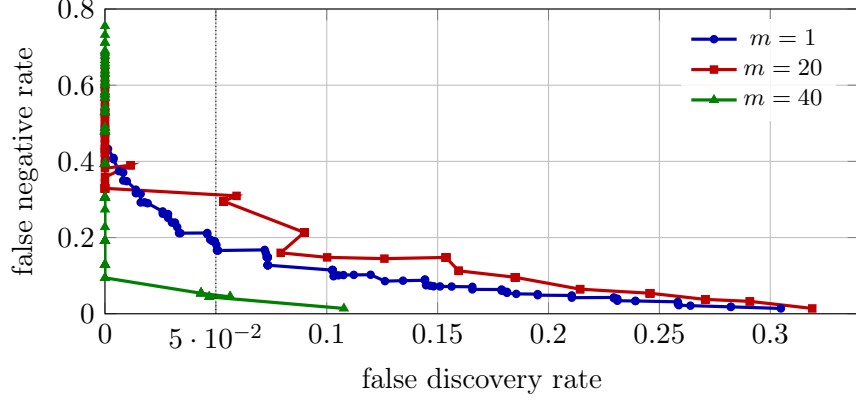


Figure 3: Rate frontiers traced out by the threshold grid for three market sizes. Apart from m , the parameter setting is the same across curves. The plot shows computed FDR against the false negative rate $1 - \text{sensitivity}(\alpha)$, with $\text{sensitivity}(\alpha) = \mathbb{E}[U(\alpha)]/\mathbb{E}[T]$; the dotted line marks the cap $\text{FDR} \leq 0.05$. Each marker is one threshold value.

Figure 4 summarizes each frontier through the principal's best threshold choice under an FDR cap. For each displayed market size, the principal searches the threshold grid and chooses the threshold that maximizes expected true approvals subject to $\text{FDR} \leq 0.05$. The optimized outcome is not monotone in a single force. Larger markets create more potential high-quality approvals, but they also reduce the fixed-point reward through payoff dilution and thereby change who enters.

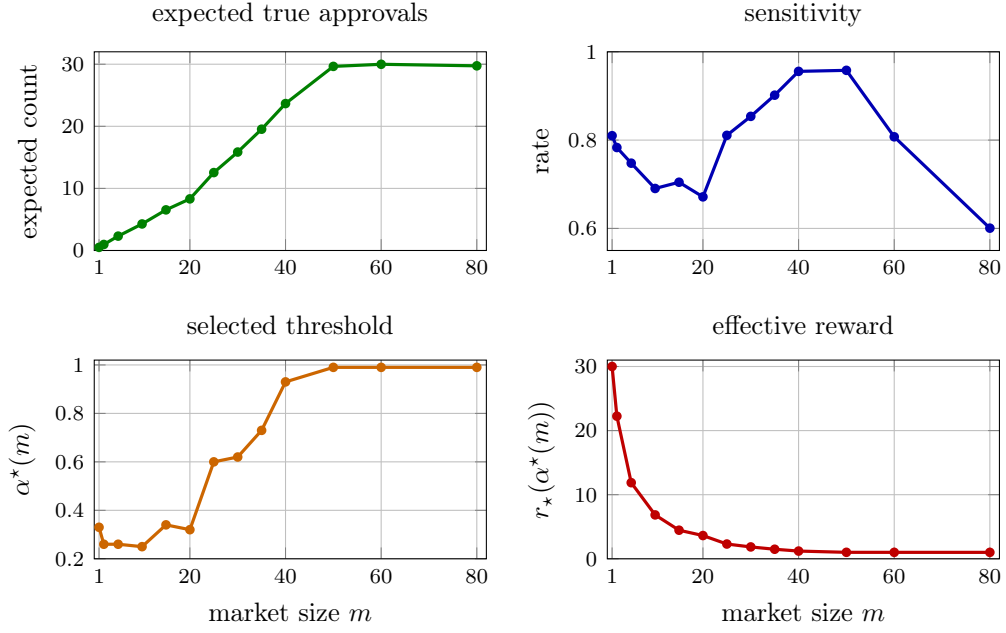


Figure 4: Best threshold choice under an FDR cap by market size. For each displayed m , the threshold $\alpha^*(m)$ maximizes expected true approvals over the grid $\{0.01, 0.02, \dots, 0.99\}$ subject to computed $\text{FDR} \leq 0.05$. The upper-left panel plots the resulting expected number of true approvals; the upper-right panel plots the corresponding sensitivity; the lower-left panel plots the selected threshold; and the lower-right panel plots the fixed-point effective reward at that threshold. Markers are the computed market sizes, and line segments are visual guides.

6.3 Competition and the baseline-entry boundary

Table 1 reports the numerical counterpart to Proposition 3. Stronger competition pushes the competition-adjusted baseline-entry boundary $\hat{\alpha}_{\text{comp}}$ to the right, while larger rewards pull it back to the left. Repeated values in the reward comparison reflect the finite threshold grid. The table should therefore be read as evidence for weak monotonicity rather than strict monotonicity.

Parameter	Value	Result for $\hat{\alpha}_{\text{comp}}$
Competition	$\gamma = 0.0$	0.10 (grid lower endpoint)
Competition	$\gamma = 0.5$	0.10 (grid lower endpoint)
Competition	$\gamma = 1.0$	0.274414
Competition	$\gamma = 1.5$	no baseline entry on $[0.10, 0.80]$
Competition	$\gamma = 2.0$	no baseline entry on $[0.10, 0.80]$
Reward	$R = 20$	0.612793
Reward	$R = 30$	0.274414
Reward	$R = 40$	0.274414
Reward	$R = 50$	0.171875
Reward	$R = 60$	0.171875

Table 1: Selected comparative statics for the competition-adjusted baseline-entry boundary $\hat{\alpha}_{\text{comp}}$ under the baseline parameter setting $m = 20$, $\mu_b = 0.5$, $q = \text{Beta}(6, 5)$, $n \in \{10, \dots, 60\}$, $c_0 = 0.5$, $c = 0.05$, and exact binomial tails. Stronger competition expands the zero-false-approval region, while larger rewards contract it.

6.4 FDR notions and upper bounds

Table 2 illustrates the relations among mFDR, pFDR, and FDR. The sparse small-market row makes the gap $\text{FDR} < \text{pFDR} = \text{mFDR}$ visible because $\mathbb{P}(S > 0)$ is materially below one. The baseline positive-error row shows the same relations in the main numerical setting, where $\mathbb{P}(S > 0)$ is almost one. The zero-false-approval row shows that the finite-grid set of thresholds with zero false approvals from Proposition 2 coexists with frequent approvals.

Setting	(m, R, α)	$\mathbb{P}(S > 0)$	pFDR = mFDR	FDR
Sparse small market	(2, 8, 0.20)	0.638764	0.014963	0.009558
Baseline positive-error	(20, 30, 0.28)	0.999934	0.011545	0.011545
Zero-false-approval baseline	(20, 30, 0.20)	0.999782	0	0

Table 2: Numerical illustration of the relations among mFDR, pFDR, and FDR in the symmetric benchmark.

Table 3 reports the hierarchy of FDR upper bounds from Theorem 1. The action-specific bound is the tightest, the null-entry bound is only slightly looser, and the coarse cost-based bound can be very loose. The rows at $\alpha = 0.28$ and $\alpha = 0.30$ also make the local reversal visible: false approvals appear at $\alpha = 0.28$ and then disappear again at $\alpha = 0.30$, while expected true approvals rise.

α	computed FDR	action-specific bound	null-entry bound	cost-based bound	true approvals
0.28	0.011545	0.011703	0.011942	0.999934	7.553673
0.30	0	0	0	0.999959	7.938423
0.38	0.059332	0.065477	0.066006	0.999995	8.548099
0.39	0.053473	0.058364	0.060384	0.999996	8.723190

Table 3: FDR upper bounds in the baseline parameter setting. Because $\mathbb{P}(S > 0)$ is nearly one on this threshold grid, the same hierarchy for mFDR is numerically almost identical to the displayed hierarchy for FDR.

6.5 Self-belief

The private-signal numerics use the same economic parameters as the baseline setting together with a Beta-Binomial pilot signal. We fix

$$\alpha = 0.05,$$

which lies in the zero-false-approval region of the known-quality benchmark, and compare known quality to private-signal equilibria with pilot sizes $k \in \{5, 20, 100\}$. Table 4 shows the main pattern from Section 5. The zero-false-approval conclusion can fail under imperfect self-knowledge: the known-quality benchmark has $\text{FDR} = 0$ at this threshold, but the self-belief model has positive false-approval probability $a_0 = \mathbb{P}(\mu_i \leq \mu_b, A_i = 1)$ and positive FDR. As the pilot signal becomes more informative, the private-signal equilibrium approaches the known-quality benchmark and the false-discovery quantities shrink rapidly.

Model	k	$r_\star$	a_0	mFDR = pFDR	FDR	cost-based bound
Known quality	–	5.9652	0	0	0	0.2983
Self-belief	5	7.0792	0.002163	0.010300	0.010208	0.3540
Self-belief	20	6.7510	0.000850	0.003852	0.003825	0.3376
Self-belief	100	6.2500	0.000031	0.000129	0.000128	0.3125

Table 4: Known quality versus self-belief at $\alpha = 0.05$ under the baseline economic parameters and a Beta-Binomial pilot stage. The known-quality benchmark has zero false approvals, while private signals can generate them; the error rates then contract toward zero as the pilot signal becomes more informative. Here $a_0 = \mathbb{P}(\mu_i \leq \mu_b, A_i = 1)$ is the representative-agent false-approval probability.

7 Discussion

This paper studies threshold design when evidence is produced by competing strategic agents. In the symmetric known-quality model, the many-agent game reduces to a scalar fixed point in the effective reward. That reduction gives a competition-adjusted baseline-entry boundary, a zero-false-approval region, and relations among mFDR, pFDR, and FDR together with computable FDR upper bounds. With private signals and correct posterior beliefs, the fixed-point reduction and FDR bounds remain, but zero false approvals need not.

The main limitation is that the model is simultaneous and uses one common approval threshold. A natural next direction is timing. In many markets, early approval can matter more than later approval, and in some settings the first successful entrant may capture most or all of the reward. Studying that timing problem would require modeling not only whether agents enter and pass, but

also when they choose to generate evidence and how the principal’s rule affects the race for early approval.

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A Proofs for Symmetric Competition with Known Qualities

The proofs are organized by main-text section. For the symmetric and self-belief results, they are written at the level of induced approval probabilities and affine within-agent payoff lines. Thus the proofs apply to any one-sided evidence model satisfying the structural conditions listed in Section 3, including the Bernoulli/binomial and one-sided normal (Gaussian) examples discussed there.

A.1 Proof details for Proposition 1

Fix a threshold α and let the admissible action menu be

$$\mathcal{N} = \{0, n_{\min}, \dots, n_{\max}\}.$$

Write

$$\mathcal{N}_+ = \mathcal{N} \setminus \{0\}$$

for the positive-action menu. For fixed quality μ and reward level r , define

$$P_n(\alpha, \mu) = \text{Pass}(\alpha, \mu, n), \quad U_n(\alpha, \mu; r) = rP_n(\alpha, \mu) - C(n),$$

with $P_0(\alpha, \mu) = C(0) = 0$. Let $\text{BR}(\alpha, r)(\mu)$ denote the selected best response under the tie-break rule from Proposition 1.

Lemma 1 (Monotone chosen pass probability). *For fixed (α, μ) , the pass probability of the selected best response is nondecreasing in r .*

Proof. Take $r_1 < r_2$ and let

$$n_1 = \text{BR}(\alpha, r_1)(\mu), \quad n_2 = \text{BR}(\alpha, r_2)(\mu).$$

By optimality,

$$U_{n_1}(\alpha, \mu; r_1) \geq U_{n_2}(\alpha, \mu; r_1), \quad U_{n_2}(\alpha, \mu; r_2) \geq U_{n_1}(\alpha, \mu; r_2).$$

Subtracting yields

$$(r_2 - r_1)(P_{n_2}(\alpha, \mu) - P_{n_1}(\alpha, \mu)) \geq 0.$$

Since $r_2 - r_1 > 0$, we obtain

$$P_{n_2}(\alpha, \mu) \geq P_{n_1}(\alpha, \mu).$$

At breakpoints where several actions have the same utility, the tie-break rule selects the one with larger pass probability. Thus the chosen pass probability cannot jump downward at breakpoints either. $\square$

Lemma 2 (Monotone aggregate approval probability). *Let*

$$\rho(\alpha, r) = \mathbb{E}_\mu [\text{Pass}(\alpha, \mu, \text{BR}(\alpha, r)(\mu))].$$

Then $\rho(\alpha, r)$ is nondecreasing in r .

Proof. Lemma 1 gives pointwise monotonicity in r for each fixed quality μ , and integrating over the quality distribution preserves that ordering. $\square$

Lemma 3 (Continuity of the aggregate approval probability). *Suppose the quality distribution is atomless and that for every reward level r and every distinct pair of actions $n \neq n'$ in $\mathcal{N}$, the indifference set*

$$\{\mu : U_n(\alpha, \mu; r) = U_{n'}(\alpha, \mu; r)\}$$

has measure zero. Then

$$r \mapsto \rho(\alpha, r)$$

is continuous on $[0, R]$.

Proof. Take a sequence $r_k \rightarrow r$ in $[0, R]$. For every quality μ outside the union of the pairwise indifference sets at r , the maximizing action at r is unique. Because the action menu is finite and each payoff line $U_n(\alpha, \mu; \cdot)$ is continuous in r , that unique maximizer remains optimal for all r_k sufficiently close to r . Hence

$$\text{BR}(\alpha, r_k)(\mu) = \text{BR}(\alpha, r)(\mu)$$

eventually for almost every μ .

Therefore

$$\text{Pass}(\alpha, \mu, \text{BR}(\alpha, r_k)(\mu)) \rightarrow \text{Pass}(\alpha, \mu, \text{BR}(\alpha, r)(\mu))$$

for almost every μ . Since $0 \leq \text{Pass} \leq 1$, dominated convergence gives

$$\rho(\alpha, r_k) \rightarrow \rho(\alpha, r).$$

$\square$

Lemma 4 (Approval dilution is antitone in the approval rate). *Fix m and a decreasing dilution function g . If $0 \leq p_1 \leq p_2 \leq 1$ and*

$$Y_\ell \sim \text{Binomial}(m-1, p_\ell), \quad \ell \in \{1, 2\},$$

then

$$\mathbb{E}[g(1 + Y_1)] \geq \mathbb{E}[g(1 + Y_2)].$$

Proof. The binomial family is increasing in the success probability in first-order stochastic dominance. Thus Y_2 dominates Y_1 . Applying the decreasing function $g(1 + \cdot)$ reverses the ordering after taking expectations. $\square$

Proof of Proposition 1. Define

$$\Phi_\alpha(r) = R \mathbb{E}[g(1 + Y_r)], \quad Y_r \sim \text{Binomial}(m - 1, \rho(\alpha, r)).$$

If all rivals use the best response induced by reward r , then their common approval probability is $\rho(\alpha, r)$. Hence the continuation reward conditional on approval is exactly $\Phi_\alpha(r)$. Hence any symmetric equilibrium must satisfy the fixed-point condition

$$r_\star(\alpha) = \Phi_\alpha(r_\star(\alpha)).$$

By Lemma 2, the aggregate approval probability $\rho(\alpha, r)$ is nondecreasing in r . Lemma 4 then implies that Φ_α is nonincreasing in r . By Lemma 3, the aggregate approval probability $\rho(\alpha, r)$ is continuous in r . For fixed m and decreasing g , the map

$$p \mapsto \mathbb{E}[g(1 + Y)], \quad Y \sim \text{Binomial}(m - 1, p),$$

is a polynomial in p , hence continuous. Therefore Φ_α is continuous on $[0, R]$.

Now define

$$F_\alpha(r) = \Phi_\alpha(r) - r.$$

Since Φ_α maps $[0, R]$ into itself, we have

$$F_\alpha(0) = \Phi_\alpha(0) \geq 0, \quad F_\alpha(R) = \Phi_\alpha(R) - R \leq 0.$$

The intermediate value theorem therefore gives at least one root in $[0, R]$.

For uniqueness, if $r_2 > r_1$, then

$$F_\alpha(r_2) - F_\alpha(r_1) = \Phi_\alpha(r_2) - \Phi_\alpha(r_1) - (r_2 - r_1) \leq -(r_2 - r_1) < 0.$$

Thus F_α is strictly decreasing and can cross zero at most once.

Finally, let $r_\star(\alpha)$ be any fixed point and suppose every agent other than agent i uses the strategy

$$\text{BR}(\alpha, r_\star(\alpha)).$$

Then each rival is approved independently with common probability

$$\rho(\alpha, r_\star(\alpha)),$$

so the number of rival approvals faced by agent i satisfies

$$Y_{-i} \sim \text{Binomial}(m - 1, \rho(\alpha, r_\star(\alpha))),$$

regardless of agent i 's own action choice. Therefore, conditional on agent i being approved, its expected continuation reward is

$$R \mathbb{E}[g(1 + Y_{-i})] = \Phi_\alpha(r_\star(\alpha)) = r_\star(\alpha).$$

Hence if a type- μ_i agent deviates to any action $n \in \mathcal{N}$, its expected payoff is exactly

$$r_\star(\alpha) \text{Pass}(\alpha, \mu_i, n) - C(n).$$

By definition,

$$\text{BR}(\alpha, r_\star(\alpha))(\mu_i)$$

maximizes this expression over $n \in \mathcal{N}$. Hence no unilateral deviation is profitable for any quality μ_i , which is exactly the symmetric equilibrium condition. The first paragraph already showed that any symmetric equilibrium induces the same fixed-point condition. $\square$

A.2 Proof details for Propositions 2 and 3

For fixed threshold α , let $r_\star(\alpha)$ denote the unique equilibrium effective reward. Recall that

$$\hat{\alpha}_{\text{comp}} = \inf\{\alpha : V^+(\alpha, \mu_b; r_\star(\alpha)) \geq 0\},$$

with $\inf \emptyset = +\infty$, where

$$V^+(\alpha, \mu; r) = \max_{n \in \mathcal{N}_+} \{rP_n(\alpha, \mu) - C(n)\}.$$

Because staying out has utility and pass probability both equal to zero, while the tie-break rule favors larger pass probability, the selected best response is positive if and only if $V^+(\alpha, \mu; r) \geq 0$.

Lemma 5 (Monotonicity of positive-entry value). *For fixed quality μ , the positive-entry value*

$$V^+(\alpha, \mu; r) = \max_{n \in \mathcal{N}_+} \{rP_n(\alpha, \mu) - C(n)\}$$

is nondecreasing in α , μ , and r .

Proof. For fixed (μ, r, n) , the pass probability $P_n(\alpha, \mu)$ is nondecreasing in α because the one-sided rejection region expands with the threshold. Thus each payoff line

$$rP_n(\alpha, \mu) - C(n)$$

is nondecreasing in α . Taking the maximum over the finite action menu preserves the ordering.

For fixed (α, r, n) , the pass probability $P_n(\alpha, \mu)$ is nondecreasing in μ by the structural conditions from Section 3. Thus each payoff line is also nondecreasing in μ , and the maximum again preserves the ordering.

For fixed (α, μ, n) , the same payoff line is affine in r with nonnegative slope $P_n(\alpha, \mu) \geq 0$. Thus it is nondecreasing in r , and again the maximum preserves the ordering. $\square$

Proof of Proposition 2. If $\alpha < \hat{\alpha}_{\text{comp}}$, then by definition of the baseline-entry boundary,

$$V^+(\alpha, \mu_b; r_\star(\alpha)) < 0.$$

Because staying out has utility and pass probability both equal to zero, while the tie-break rule favors larger pass probability, this means the baseline-quality agent strictly prefers staying out.

Lemma 5 shows that for fixed $(\alpha, r_\star(\alpha))$, the positive-entry value is nondecreasing in μ . Therefore every quality $\mu \leq \mu_b$ also satisfies

$$V^+(\alpha, \mu; r_\star(\alpha)) \leq V^+(\alpha, \mu_b; r_\star(\alpha)) < 0,$$

so all null-quality agents stay out as well. Hence no null-quality agent can be approved, the false-approval count is almost surely zero, and the zero-denominator convention from Section 3 gives

$$\text{mFDR}(\alpha) = \text{pFDR}(\alpha) = \text{FDR}(\alpha) = 0.$$

$\square$

Lemma 6 (Monotone comparison of equilibrium rewards). *Fix α and suppose the unique fixed point exists under each environment.*

(i) *If $g_{\text{lo}}(k) \leq g_{\text{hi}}(k)$ for all k , then*

$$r_\star^{\text{lo}}(\alpha) \leq r_\star^{\text{hi}}(\alpha).$$

(ii) If the monopoly reward increases from R_1 to $R_2 \geq R_1$, then

$$r_{\star, R_1}(\alpha) \leq r_{\star, R_2}(\alpha).$$

Proof. For part (i), the pointwise ordering of the dilution functions implies

$$\Phi_{\alpha}^{\text{lo}}(r) \leq \Phi_{\alpha}^{\text{hi}}(r)$$

for every r . Evaluating at $r = r_{\star}^{\text{lo}}(\alpha)$ gives

$$r_{\star}^{\text{lo}}(\alpha) = \Phi_{\alpha}^{\text{lo}}(r_{\star}^{\text{lo}}(\alpha)) \leq \Phi_{\alpha}^{\text{hi}}(r_{\star}^{\text{lo}}(\alpha)).$$

Since $r \mapsto \Phi_{\alpha}^{\text{hi}}(r) - r$ is strictly decreasing by the proof of Proposition 1, its unique zero must lie weakly to the right of $r_{\star}^{\text{lo}}(\alpha)$.

Part (ii) is identical: if $R_2 \geq R_1$, then

$$\Phi_{\alpha, R_2}(r) \geq \Phi_{\alpha, R_1}(r)$$

for every r , and the same comparison argument gives

$$r_{\star, R_1}(\alpha) \leq r_{\star, R_2}(\alpha).$$

□

Proof of Proposition 3. For each environment, let

$$E = \{\alpha : V^+(\alpha, \mu_b; r_{\star}(\alpha)) \geq 0\}$$

be the baseline-entry set under the stated tie-break rule. Then $\hat{\alpha}_{\text{comp}} = \inf E$.

For part (i), Lemma 6 gives

$$r_{\star}^{\text{lo}}(\alpha) \leq r_{\star}^{\text{hi}}(\alpha)$$

for every α . Lemma 5 then implies

$$V^+(\alpha, \mu_b; r_{\star}^{\text{lo}}(\alpha)) \leq V^+(\alpha, \mu_b; r_{\star}^{\text{hi}}(\alpha)).$$

Thus if α belongs to the low-reward entry set, it also belongs to the high-reward entry set. Therefore

$$E_{\text{lo}} \subseteq E_{\text{hi}},$$

and taking infima yields

$$\hat{\alpha}_{\text{comp}}(g_{\text{lo}}) \geq \hat{\alpha}_{\text{comp}}(g_{\text{hi}}).$$

Part (ii) is the same argument with the reward comparison

$$r_{\star, R_1}(\alpha) \leq r_{\star, R_2}(\alpha),$$

which gives

$$E_{R_1} \subseteq E_{R_2} \quad \text{and hence} \quad \hat{\alpha}_{\text{comp}}(R_2) \leq \hat{\alpha}_{\text{comp}}(R_1).$$

□

A.3 Proof details for Theorem 1

For this subsection, let

$$Z_i = \mathbf{1}\{\mu_i \leq \mu_b\}, \quad V = \sum_{i=1}^m Z_i A_i, \quad S = \sum_{i=1}^m A_i.$$

Under a symmetric equilibrium, the pairs (Z_i, A_i) are i.i.d. across agents. Write

$$\begin{aligned} \rho(\alpha) &= \mathbb{P}(A_i = 1), & a_0(\alpha) &= \mathbb{P}(Z_i = 1, A_i = 1). \\ \nu_n(\alpha) &= \mathbb{P}(\mu_i \leq \mu_b, n_i = n), & \nu(\alpha) &= \sum_{n>0} \nu_n(\alpha). \end{aligned}$$

Let

$$n_i = \text{BR}(\alpha, r_\star(\alpha))(\mu_i), \quad P_i = \text{Pass}(\alpha, \mu_i, n_i).$$

For each positive sample size n , define

$$B_{i,n}^0 = \mathbf{1}\{\mu_i \leq \mu_b, n_i = n\}, \quad \bar{\alpha}_n(\alpha) = \sup_{\mu \leq \mu_b} \text{Pass}(\alpha, \mu, n) = \text{Pass}(\alpha, \mu_b, n).$$

Since the action choice is a measurable function of μ_i alone in the known-quality model, the formal trial randomness is conditionally independent after conditioning on μ_i , and therefore

$$\mathbb{E}[A_i \mid \mu_i] = P_i.$$

Lemma 7 (Action-specific and null-entry bounds). *For the known-quality symmetric equilibrium,*

$$a_0(\alpha) \leq \sum_{n>0} \bar{\alpha}_n(\alpha) \nu_n(\alpha) \leq \alpha \nu(\alpha).$$

Proof. Since n_i takes values in the positive action menu or 0, we can write

$$Z_i P_i = \sum_{n>0} \mathbf{1}\{\mu_i \leq \mu_b, n_i = n\} \text{Pass}(\alpha, \mu_i, n).$$

For each fixed $n > 0$, monotonicity in μ and null validity imply

$$\text{Pass}(\alpha, \mu_i, n) \leq \bar{\alpha}_n(\alpha) \leq \alpha \quad \text{whenever } \mu_i \leq \mu_b.$$

Therefore

$$Z_i P_i \leq \sum_{n>0} \bar{\alpha}_n(\alpha) B_{i,n}^0 \leq \alpha \sum_{n>0} B_{i,n}^0 \quad \text{almost surely.}$$

Taking expectations and using

$$\mathbb{E}[Z_i A_i] = \mathbb{E}[Z_i P_i], \quad \mathbb{E}[B_{i,n}^0] = \nu_n(\alpha), \quad \sum_{n>0} \mathbb{E}[B_{i,n}^0] = \nu(\alpha),$$

gives the claimed inequalities. □

Lemma 8 (Cost-based comparison for null entry). *For the known-quality symmetric equilibrium,*

$$\nu(\alpha) \leq \frac{r_\star(\alpha)}{C_{\min}} \rho(\alpha).$$

Proof. Let

$$B_i = \mathbf{1}\{n_i > 0\}$$

be the entry indicator. Because staying out is feasible, the selected action weakly maximizes utility against the fixed reward $r_\star(\alpha)$, so

$$r_\star(\alpha)P_i - C(n_i) \geq 0 \quad \text{almost surely.}$$

If $n_i = 0$, then both sides are zero. If $n_i > 0$, then $C(n_i) \geq C_{\min}$. Hence

$$C_{\min}B_i \leq C(n_i) \leq r_\star(\alpha)P_i \quad \text{almost surely,}$$

which gives

$$B_i \leq \frac{r_\star(\alpha)}{C_{\min}}P_i.$$

Since null entry is a subset of total entry,

$$\sum_{n>0} B_{i,n}^0 = \mathbf{1}\{\mu_i \leq \mu_b, n_i > 0\} \leq B_i.$$

Taking expectations yields

$$\nu(\alpha) \leq \mathbb{E}[B_i] \leq \frac{r_\star(\alpha)}{C_{\min}}\mathbb{E}[P_i] = \frac{r_\star(\alpha)}{C_{\min}}\rho(\alpha).$$

□

Lemma 9 (Homogeneous FDR bound). *For the known-quality symmetric equilibrium,*

$$\mathbb{E}[V] \leq \alpha \frac{r_\star(\alpha)}{C_{\min}}\mathbb{E}[S].$$

Proof. Lemma 7 gives

$$a_0(\alpha) \leq \alpha \nu(\alpha),$$

and Lemma 8 gives

$$\nu(\alpha) \leq \frac{r_\star(\alpha)}{C_{\min}}\rho(\alpha).$$

Combining the two displays yields

$$a_0(\alpha) \leq \alpha \frac{r_\star(\alpha)}{C_{\min}}\rho(\alpha).$$

Since

$$\mathbb{E}[V] = m a_0(\alpha), \quad \mathbb{E}[S] = m \rho(\alpha),$$

the claimed bound follows. □

Lemma 10 (Reciprocal binomial formula). *If $Y \sim \text{Binomial}(m-1, \rho)$ with $\rho \in (0, 1]$, then*

$$\mathbb{E}\left[\frac{1}{1+Y}\right] = \frac{1 - (1-\rho)^m}{m\rho}.$$

Proof. Expand the expectation and use

$$\frac{1}{k+1} \binom{m-1}{k} = \frac{1}{m} \binom{m}{k+1}.$$

This gives

$$\mathbb{E} \left[\frac{1}{1+Y} \right] = \frac{1}{m\rho} \sum_{j=1}^m \binom{m}{j} \rho^j (1-\rho)^{m-j} = \frac{1 - (1-\rho)^m}{m\rho}.$$

□

Lemma 11 (Relationships among mFDR, pFDR, and FDR). *If the pairs*

$$(Z_i, A_i) = (\mathbf{1}\{\mu_i \leq \mu_b\}, A_i)$$

are i.i.d. across agents and A_i has common success probability $\rho(\alpha)$, then

$$\text{pFDR}(\alpha) = \text{mFDR}(\alpha), \quad \text{FDR}(\alpha) = \mathbb{P}(S(\alpha) > 0) \text{mFDR}(\alpha).$$

Proof. If $\rho(\alpha) = 0$, then $S(\alpha) = 0$ almost surely. The convention from Section 3 gives

$$\text{mFDR}(\alpha) = \text{pFDR}(\alpha) = \text{FDR}(\alpha) = 0.$$

Now suppose $\rho(\alpha) > 0$.

By exchangeability,

$$\mathbb{E}[V] = ma_0(\alpha), \quad \mathbb{E}[S] = m\rho(\alpha),$$

so

$$\text{mFDR}(\alpha) = \frac{a_0(\alpha)}{\rho(\alpha)}.$$

For FDR, let

$$S_{-i} = \sum_{j \neq i} A_j.$$

Since $Z_i A_i = 0$ whenever $A_i = 0$,

$$\frac{V}{\max(S, 1)} = \sum_{i=1}^m \frac{Z_i A_i}{1 + S_{-i}}.$$

Independence across agents implies

$$S_{-i} \sim \text{Binomial}(m-1, \rho(\alpha))$$

and is independent of (Z_i, A_i) . Therefore

$$\text{FDR}(\alpha) = \sum_{i=1}^m \mathbb{E}[Z_i A_i] \mathbb{E} \left[\frac{1}{1 + S_{-i}} \right] = ma_0(\alpha) \mathbb{E} \left[\frac{1}{1 + Y} \right],$$

where $Y \sim \text{Binomial}(m-1, \rho(\alpha))$. Applying Lemma 10,

$$\text{FDR}(\alpha) = (1 - (1 - \rho(\alpha))^m) \frac{a_0(\alpha)}{\rho(\alpha)} = \mathbb{P}(S(\alpha) > 0) \text{mFDR}(\alpha).$$

Since by definition

$$\text{FDR}(\alpha) = \mathbb{P}(S(\alpha) > 0) \text{pFDR}(\alpha),$$

we obtain

$$\text{pFDR}(\alpha) = \text{mFDR}(\alpha).$$

□

Proof of Theorem 1. If $\rho(\alpha) = 0$, then $\mathbb{E}[S] = m\rho(\alpha) = 0$. The convention from Section 3 gives

$$\text{mFDR}(\alpha) = 0.$$

Since $\mathbb{E}[S] = 0$ implies $S(\alpha) = 0$ almost surely, the same convention also gives

$$\text{pFDR}(\alpha) = \text{FDR}(\alpha) = 0.$$

The claimed bound and relationships are therefore immediate.

Now suppose $\rho(\alpha) > 0$. Then $\mathbb{E}[S] = m\rho(\alpha) > 0$, and Lemma 7 gives

$$a_0(\alpha) \leq \sum_{n>0} \bar{\alpha}_n(\alpha) \nu_n(\alpha) \leq \alpha \nu(\alpha).$$

Dividing by $\rho(\alpha)$ yields

$$\text{mFDR}(\alpha) = \frac{a_0(\alpha)}{\rho(\alpha)} \leq \frac{\sum_{n>0} \bar{\alpha}_n(\alpha) \nu_n(\alpha)}{\rho(\alpha)} \leq \alpha \frac{\nu(\alpha)}{\rho(\alpha)}.$$

Lemma 8 then gives

$$\alpha \frac{\nu(\alpha)}{\rho(\alpha)} \leq \alpha \frac{r_\star(\alpha)}{C_{\min}}.$$

Combining the last two displays gives the full hierarchy of mFDR bounds.

Lemma 11 gives the relationships

$$\text{pFDR}(\alpha) = \text{mFDR}(\alpha), \quad \text{FDR}(\alpha) = \mathbb{P}(S(\alpha) > 0) \text{mFDR}(\alpha).$$

Combining these displays yields

$$\text{FDR}(\alpha) \leq \mathbb{P}(S(\alpha) > 0) \frac{\sum_{n>0} \bar{\alpha}_n(\alpha) \nu_n(\alpha)}{\rho(\alpha)} \leq \mathbb{P}(S(\alpha) > 0) \alpha \frac{\nu(\alpha)}{\rho(\alpha)} \leq \alpha \frac{r_\star(\alpha)}{C_{\min}}.$$

If $\alpha < \hat{\alpha}_{\text{comp}}$, Proposition 2 gives

$$\text{mFDR}(\alpha) = \text{pFDR}(\alpha) = \text{FDR}(\alpha) = 0.$$

□

B Proofs for Private Signals and Self-Belief

B.1 Proof details for Proposition 4, Theorem 2, and Proposition 5

Fix a threshold α , a reward level r , an ordered signal space $\mathcal{S}$, and a finite action menu

$$\mathcal{A} = \{0\} \cup \mathcal{N}.$$

For a signal value $s \in \mathcal{S}$, let

$$\Pi_s = \text{Law}(\mu_i \mid S_i = s)$$

denote the correct posterior belief about the agent's own quality. For each action $a \in \mathcal{A}$, define the posterior expected pass probability and posterior expected utility by

$$P_{\text{SB}}(\alpha, a; s) = \mathbb{E}_{\mu \sim \Pi_s}[\text{Pass}(\alpha, \mu, a)],$$

$$U_{\text{SB}}(\alpha, r, a; s) = r P_{\text{SB}}(\alpha, a; s) - C(a).$$

Let $\text{BR}_{\text{SB}}(\alpha, r)(s)$ denote the selected best response under the tie-break rule from Theorem 2.

Lemma 12 (Posterior expected pass probability is monotone in signal). *If $s' \geq s$ implies that $\Pi_{s'}$ first-order stochastically dominates Π_s , then for each fixed (α, a) ,*

$$P_{\text{SB}}(\alpha, a; s') \geq P_{\text{SB}}(\alpha, a; s).$$

Proof. For fixed (α, a) , the map

$$\mu \mapsto \text{Pass}(\alpha, \mu, a)$$

is bounded and nondecreasing by the structural conditions from Section 3. First-order stochastic dominance of the posterior family therefore implies the same ordering after integration. $\square$

Proof of Proposition 4. Lemma 12 shows that posterior expected pass probability is nondecreasing in the signal for each fixed action. Multiplying by $r \geq 0$ and subtracting the action cost preserves the ordering. Thus each action-specific utility is nondecreasing in s . Taking the maximum over the finite action menu preserves monotonicity of the value function and therefore makes the entry set an upper set. $\square$

Lemma 13 (Chosen posterior expected pass probability is monotone in reward). *Fix (α, s) . Under the tie-break rule that favors the larger posterior expected pass probability and then the smaller positive action, the chosen posterior expected pass probability*

$$P_{\text{BR,SB}}(\alpha, s; r) = P_{\text{SB}}(\alpha, \text{BR}_{\text{SB}}(\alpha, r)(s); s)$$

is nondecreasing in r .

Proof. For fixed (α, s) , each action induces an affine payoff line

$$\ell_a(r) = r P_{\text{SB}}(\alpha, a; s) - C(a)$$

with slope equal to the posterior expected pass probability. The maximum of finitely many affine functions is convex. Therefore the slope of the selected supporting line can only move upward as r increases. The tie-break rule selects the line with the larger slope whenever several lines meet. $\square$

Lemma 14 (Continuity of the self-belief approval probability). *Suppose the signal distribution is atomless and that for every reward level r and every distinct pair of actions $a \neq a'$ in $\mathcal{A}$, the indifference set*

$$\{s : U_{\text{SB}}(\alpha, r, a; s) = U_{\text{SB}}(\alpha, r, a'; s)\}$$

has measure zero under the signal law. Then

$$r \mapsto \rho_{\text{SB}}(\alpha, r)$$

is continuous on $[0, R]$.

Proof. Take a sequence $r_k \rightarrow r$ in $[0, R]$. For every signal value s outside the union of the pairwise indifference sets at r , the maximizing action at r is unique. Because the action menu is finite and each payoff line

$$U_{\text{SB}}(\alpha, \cdot, a; s)$$

is continuous in r , that unique maximizer remains optimal for all r_k sufficiently close to r . Hence

$$\text{BR}_{\text{SB}}(\alpha, r_k)(s) = \text{BR}_{\text{SB}}(\alpha, r)(s)$$

eventually for almost every s .

Therefore

$$P_{\text{SB}}(\alpha, \text{BR}_{\text{SB}}(\alpha, r_k)(s); s) \rightarrow P_{\text{SB}}(\alpha, \text{BR}_{\text{SB}}(\alpha, r)(s); s)$$

for almost every s . Since $0 \leq P_{\text{SB}} \leq 1$, dominated convergence gives

$$\mathbb{E}[P_{\text{SB}}(\alpha, \text{BR}_{\text{SB}}(\alpha, r_k)(S_i); S_i)] \rightarrow \mathbb{E}[P_{\text{SB}}(\alpha, \text{BR}_{\text{SB}}(\alpha, r)(S_i); S_i)].$$

By iterated expectations,

$$\rho_{\text{SB}}(\alpha, r) = \mathbb{E}[P_{\text{SB}}(\alpha, \text{BR}_{\text{SB}}(\alpha, r)(S_i); S_i)],$$

so $\rho_{\text{SB}}(\alpha, r_k) \rightarrow \rho_{\text{SB}}(\alpha, r)$. □

Lemma 15 (Self-belief action-specific and null-entry bounds). *Let $r_{\text{SB}}^*(\alpha)$ be the unique self-belief fixed point, and define*

$$n_i = \text{BR}_{\text{SB}}(\alpha, r_{\text{SB}}^*(\alpha))(S_i), \quad a_{0,\text{SB}}(\alpha) = \mathbb{P}(\mu_i \leq \mu_b, A_i = 1).$$

For each $n > 0$, define

$$\nu_{\text{SB},n}(\alpha) = \mathbb{P}(\mu_i \leq \mu_b, n_i = n), \quad \nu_{\text{SB}}(\alpha) = \sum_{n>0} \nu_{\text{SB},n}(\alpha),$$

and

$$\bar{\alpha}_n(\alpha) = \sup_{\mu \leq \mu_b} \text{Pass}(\alpha, \mu, n) = \text{Pass}(\alpha, \mu_b, n).$$

Then

$$a_{0,\text{SB}}(\alpha) \leq \sum_{n>0} \bar{\alpha}_n(\alpha) \nu_{\text{SB},n}(\alpha) \leq \alpha \nu_{\text{SB}}(\alpha).$$

Proof. For each agent,

$$\mathbf{1}\{\mu_i \leq \mu_b\} \text{Pass}(\alpha, \mu_i, n_i) \leq \sum_{n>0} \bar{\alpha}_n(\alpha) \mathbf{1}\{\mu_i \leq \mu_b, n_i = n\}.$$

This uses only that n_i is chosen before the formal trial data are observed, that the chosen action is measurable with respect to the private signal, and that one-sided validity plus monotonicity in μ imply

$$\text{Pass}(\alpha, \mu_i, n) \leq \bar{\alpha}_n(\alpha) \leq \alpha \quad \text{whenever } \mu_i \leq \mu_b.$$

Taking expectations and using

$$\mathbb{E}[A_i \mid \mu_i, S_i] = \text{Pass}(\alpha, \mu_i, n_i)$$

gives

$$a_{0,\text{SB}}(\alpha) \leq \sum_{n>0} \bar{\alpha}_n(\alpha) \nu_{\text{SB},n}(\alpha) \leq \alpha \nu_{\text{SB}}(\alpha). \quad \square$$

Lemma 16 (Self-belief cost-based comparison for null entry). *Let*

$$\rho_{\text{SB}}(\alpha) = \rho_{\text{SB}}(\alpha, r_{\text{SB}}^*(\alpha)).$$

Then

$$\nu_{\text{SB}}(\alpha) \leq \frac{r_{\text{SB}}^*(\alpha)}{C_{\min}} \rho_{\text{SB}}(\alpha).$$

Proof. Let

$$I_i = \mathbf{1}\{n_i > 0\}, \quad W_i = Rg(1 + K_{-i}), \quad C_i = C(n_i).$$

If $I_i = 1$, then $C_i \geq C_{\min}$. Therefore

$$\mathbb{E}[C_i] \geq C_{\min}\mathbb{E}[I_i].$$

Under symmetry, W_i depends only on rival approvals and is independent of agent i 's own tuple (μ_i, S_i, X_i) . By the fixed-point definition,

$$\mathbb{E}[W_i] = r_{\text{SB}}^*(\alpha), \quad \mathbb{E}[W_i A_i] = r_{\text{SB}}^*(\alpha)\rho_{\text{SB}}(\alpha).$$

Because the posterior is correct, conditional on $S_i = s$ the agent's posterior objective is its true conditional expected profit. Since staying out is feasible and yields zero, optimality implies

$$\mathbb{E}[W_i A_i - C_i \mid S_i = s] \geq 0.$$

Averaging over s gives

$$r_{\text{SB}}^*(\alpha)\rho_{\text{SB}}(\alpha) \geq \mathbb{E}[C_i] \geq C_{\min}\mathbb{E}[I_i].$$

Since null entry is a subset of total entry,

$$\nu_{\text{SB}}(\alpha) = \mathbb{P}(\mu_i \leq \mu_b, n_i > 0) \leq \mathbb{E}[I_i].$$

Combining the last two displays yields the claim. $\square$

Proof of Theorem 2. Define

$$\rho_{\text{SB}}(\alpha, r) = \mathbb{E}[\text{Pass}(\alpha, \mu_i, \text{BR}_{\text{SB}}(\alpha, r)(S_i))].$$

Define the self-belief fixed-point map

$$\Phi_{\text{SB},\alpha}(r) = R\mathbb{E}[g(1 + Y_r)], \quad Y_r \sim \text{Binomial}(m - 1, \rho_{\text{SB}}(\alpha, r)),$$

If all rivals use the signal strategy

$$\text{BR}_{\text{SB}}(\alpha, r),$$

then their common approval probability is $\rho_{\text{SB}}(\alpha, r)$, so the continuation reward conditional on approval is exactly

$$\Phi_{\text{SB},\alpha}(r).$$

Hence any symmetric equilibrium must satisfy the fixed-point condition

$$r_{\text{SB}}^*(\alpha) = \Phi_{\text{SB},\alpha}(r_{\text{SB}}^*(\alpha)).$$

By Lemma 13, for each signal value s the chosen posterior expected pass probability is non-decreasing in r . Integrating over the signal distribution implies that $\rho_{\text{SB}}(\alpha, r)$ is nondecreasing in r . Lemma 4 then implies that $\Phi_{\text{SB},\alpha}$ is nonincreasing in r . By Lemma 14, the self-belief approval probability $\rho_{\text{SB}}(\alpha, r)$ is continuous in r . For fixed m and decreasing g , the map

$$p \mapsto \mathbb{E}[g(1 + Y)], \quad Y \sim \text{Binomial}(m - 1, p),$$

is a polynomial in p , hence continuous. Therefore $\Phi_{\text{SB},\alpha}$ is continuous on $[0, R]$.

Now define

$$F_{\text{SB},\alpha}(r) = \Phi_{\text{SB},\alpha}(r) - r.$$

Since $\Phi_{\text{SB},\alpha}$ maps $[0, R]$ into itself, we have

$$F_{\text{SB},\alpha}(0) = \Phi_{\text{SB},\alpha}(0) \geq 0, \quad F_{\text{SB},\alpha}(R) = \Phi_{\text{SB},\alpha}(R) - R \leq 0.$$

The intermediate value theorem therefore gives at least one root in $[0, R]$.

For uniqueness, if $r_2 > r_1$, then

$$F_{\text{SB},\alpha}(r_2) - F_{\text{SB},\alpha}(r_1) = \Phi_{\text{SB},\alpha}(r_2) - \Phi_{\text{SB},\alpha}(r_1) - (r_2 - r_1) \leq -(r_2 - r_1) < 0.$$

Thus $F_{\text{SB},\alpha}$ is strictly decreasing and can cross zero at most once. Let $r_{\text{SB}}^*(\alpha)$ denote that unique root. If every agent other than agent i uses the strategy

$$\text{BR}_{\text{SB}}(\alpha, r_{\text{SB}}^*(\alpha)),$$

then the number of rival approvals faced by agent i satisfies

$$Y_{-i} \sim \text{Binomial}(m-1, \rho_{\text{SB}}(\alpha, r_{\text{SB}}^*(\alpha))),$$

regardless of agent i 's own action choice. Therefore the continuation reward conditional on approval is

$$R\mathbb{E}[g(1 + Y_{-i})] = \Phi_{\text{SB},\alpha}(r_{\text{SB}}^*(\alpha)) = r_{\text{SB}}^*(\alpha).$$

For any signal value s and any action $a \in \mathcal{A}$, the correct posterior then gives expected payoff

$$U_{\text{SB}}(\alpha, r_{\text{SB}}^*(\alpha), a; s) = r_{\text{SB}}^*(\alpha)P_{\text{SB}}(\alpha, a; s) - C(a).$$

By definition,

$$\text{BR}_{\text{SB}}(\alpha, r_{\text{SB}}^*(\alpha))(s)$$

maximizes this expression over $a \in \mathcal{A}$. Hence no unilateral deviation is profitable for any signal value s , and the fixed point induces a symmetric equilibrium. Therefore

$$r_{\text{SB}}^*(\alpha) = \Phi_{\text{SB},\alpha}(r_{\text{SB}}^*(\alpha))$$

is the unique symmetric equilibrium reward level.

Write

$$\rho_{\text{SB}}(\alpha) = \rho_{\text{SB}}(\alpha, r_{\text{SB}}^*(\alpha)).$$

If $\rho_{\text{SB}}(\alpha) = 0$, then the zero-denominator convention from Section 3 gives

$$\text{mFDR}(\alpha) = \text{pFDR}(\alpha) = \text{FDR}(\alpha) = 0.$$

The claimed bounds and relationships are therefore immediate.

Now suppose $\rho_{\text{SB}}(\alpha) > 0$. Lemma 15 gives

$$a_{0,\text{SB}}(\alpha) \leq \sum_{n>0} \bar{\alpha}_n(\alpha) \nu_{\text{SB},n}(\alpha) \leq \alpha \nu_{\text{SB}}(\alpha),$$

so

$$\text{mFDR}(\alpha) = \frac{a_{0,\text{SB}}(\alpha)}{\rho_{\text{SB}}(\alpha)} \leq \frac{\sum_{n>0} \bar{\alpha}_n(\alpha) \nu_{\text{SB},n}(\alpha)}{\rho_{\text{SB}}(\alpha)} \leq \alpha \frac{\nu_{\text{SB}}(\alpha)}{\rho_{\text{SB}}(\alpha)}.$$

Lemma 16 then yields

$$\alpha \frac{\nu_{\text{SB}}(\alpha)}{\rho_{\text{SB}}(\alpha)} \leq \alpha \frac{r_{\text{SB}}^*(\alpha)}{C_{\min}}.$$

Under the i.i.d. self-belief environment, each approval indicator is a measurable function of the independent tuple (μ_i, S_i, X_i) . Since

$$Z_i = \mathbf{1}\{\mu_i \leq \mu_b\}$$

is also a measurable function of that same tuple, the pairs (Z_i, A_i) remain i.i.d. across agents. Lemma 11 therefore gives

$$\text{pFDR}(\alpha) = \text{mFDR}(\alpha), \quad \text{FDR}(\alpha) = \mathbb{P}(S(\alpha) > 0) \text{mFDR}(\alpha).$$

□

Proof of Proposition 5. For each fixed action a , the posterior utility is

$$U_{\text{SB}}^\eta(\alpha, r, a; S_i^\eta) = r \mathbb{E}_{\mu \sim \Pi_{\eta, S_i^\eta}}[\text{Pass}(\alpha, \mu, a)] - C(a).$$

By continuity of $\mu \mapsto \text{Pass}(\alpha, \mu, a)$ and posterior concentration at μ_i , the Portmanteau theorem implies

$$\mathbb{E}_{\mu \sim \Pi_{\eta, S_i^\eta}}[\text{Pass}(\alpha, \mu, a)] \rightarrow \text{Pass}(\alpha, \mu_i, a)$$

almost surely. Subtracting the deterministic cost gives convergence of the action-specific utilities.

Because the action menu is finite, the optimized value converges as well. If the known-quality problem has a strict optimality gap, then eventually the same action remains optimal. The same gap argument yields eventual agreement of entry decisions away from the known-quality entry boundary. □